2.2.6 Diborane

Preparation:

As discussed earlier diborane can be prepared by the action of metal hydride with boron. This method is used for the industrial production.

Diborane can also be obtained in small quantities by the reaction of iodine with sodium borohydride in diglyme.

$$2NaBH_4 + I_2 \longrightarrow B_2H_6 + 2NaI + H_2$$

On heating magnesium boride with HCl a mixture of volatile boranes are obtained.

$$2Mg_3B_2 + 12HCl \longrightarrow 6MgCl_2 + B_4H_{10} + H_2$$

$$B_4H_{10} + H_2 \longrightarrow 2B_2H_6$$

Properties:

Boranes are colourless diamagnetic compounds with low thermal stability. Diborane is a gas at room temperature with sweet smell and it is extremely toxic. It is also highly reactive.

At high temperatures it forms higher boranes liberating hydrogen.

$$5B_2H_6 \xrightarrow[\text{U - tube}]{388\ K} 2B_5H_{11} + 4H_2$$

$$2B_2H_6 \xrightarrow{198 - 373\ K} B_4H_{10} + H_2$$

$$5B_2H_6 \xrightarrow[\text{sealed tube}]{373\ K} B_{10}H_{14} + 8H_2$$

$$5B_2H_6 \xrightarrow{473 - 523\ K} 2B_5H_9 + 6H_2$$

$$10B_2H_6 \xrightarrow{523\ K} 2B_5H_9 + 2B_5H_{10} + 11H_2$$

$$B_2H_6 \xrightarrow{\text{Red hot}} 2B + 3H_2$$

Diboranes reacts with water and alkali to give boric acid and metaborates respectively.

$$B_2H_6 + 6H_2O \longrightarrow 2H_3BO_3 + 6H_2$$

$$B_2H_6 + 2NaOH + 2H_2O \longrightarrow 2NaBO_2 + 6H_2$$

Action of air:

At room temperature pure diborane does not react with air or oxygen but in impure form it gives B_2O_3 along with large amount of heat.

$$B_2H_6 + 3O_2 \longrightarrow B_2O_3 + 3H_2O \quad \Delta H = -2165\ KJ\ mol^{-1}$$

Diborane reacts with methyl alcohol to give trimethyl Borate.

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